

Lanternfishes and relatives

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Scolecophompha, Cyclosquamata
ORDERS	2
FAMILIES	18
SPECIES	About 520

Lanternfishes share with dragonfishes (see opposite) the ability to emit light from organs called photophores on their head and body. They are sometimes regarded as the deep-water equivalent of herrings because they are small, swim in large, dense schools, and form a vital element of deep-water ecosystems by providing food for many other fishes, and marine

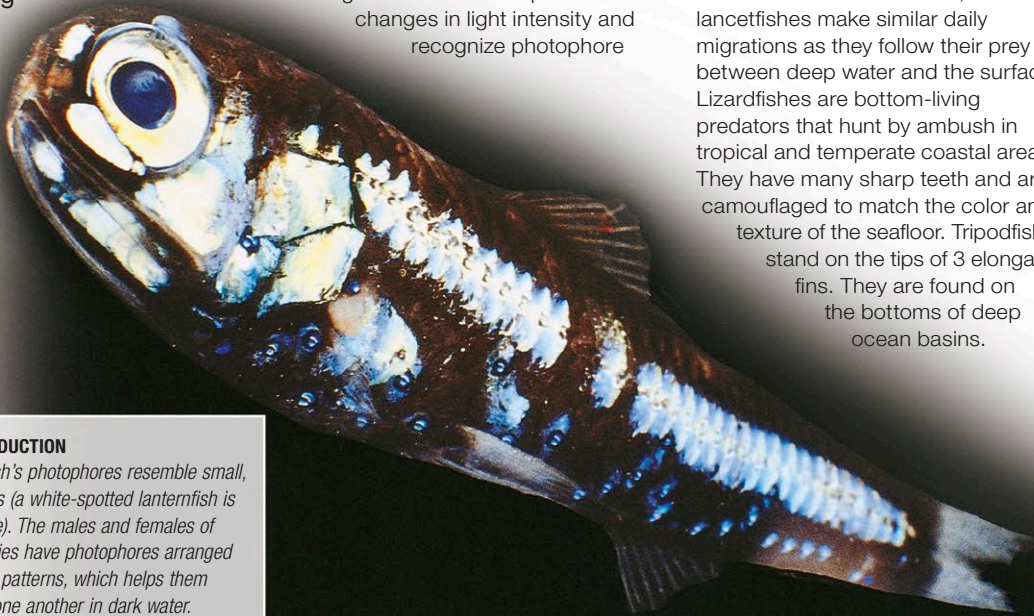
mammals. Lanternfishes are found in great numbers in the middle layers of deep waters off the continental shelves. Their relatives include lancetfish and tripodfish, which live in deep water, as well as coastal lizardfish, which use their fins to rest propped up on the ocean floor.

Anatomy

All the fishes in this group have thin scales—some silvery and others darkly pigmented—that often fall off when they are caught. Lanternfishes have bioluminescent photophores on their flanks, undersides, and heads, which emit light in shades of green, yellow,

LIGHT PRODUCTION

A lanternfish's photophores resemble small, bright studs (a white-spotted lanternfish is shown here). The males and females of some species have photophores arranged in different patterns, which helps them recognize one another in dark water.



or blue. Each species has a unique pattern of photophores, which helps groups maintain cohesion in darkly lit waters. Lanternfishes often make vertical migrations from deep water to surface waters at night to feed on plankton, returning to deep water as the sun rises. They have large eyes and good vision that help them detect changes in light intensity and recognize photophore patterns in dark water. The other fishes in this group are quite different from lanternfishes in anatomy and behavior. Most have no photophores. Lancetfishes are slender deep-water fishes. They have a large mouth containing sharp, pointed teeth and feed on other deep-water organisms. These include lanternfishes, and the lancetfishes make similar daily migrations as they follow their prey between deep water and the surface. Lizardfishes are bottom-living predators that hunt by ambush in tropical and temperate coastal areas. They have many sharp teeth and are camouflaged to match the color and texture of the seafloor. Tripodfishes stand on the tips of 3 elongated fins. They are found on the bottoms of deep ocean basins.

Bathypterois grallator

Tripodfish

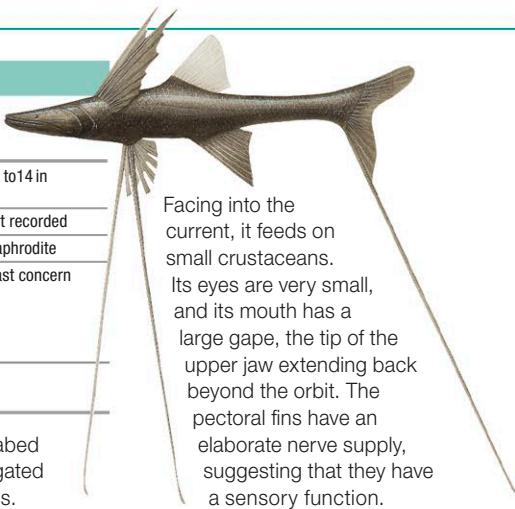


Location Atlantic, Mediterranean, Indian Ocean, W. Pacific

Length Up to 14 in (36 cm)
Weight Not recorded
Sex Hermaphrodite
Status Least concern



This fish perches over the seabed by forming a tripod with elongated rays from its pelvic and tail fins.



Facing into the current, it feeds on small crustaceans. Its eyes are very small, and its mouth has a large gape, the tip of the upper jaw extending back beyond the orbit. The pectoral fins have an elaborate nerve supply, suggesting that they have a sensory function.

Alepisaurus ferox

Long-snouted lancetfish

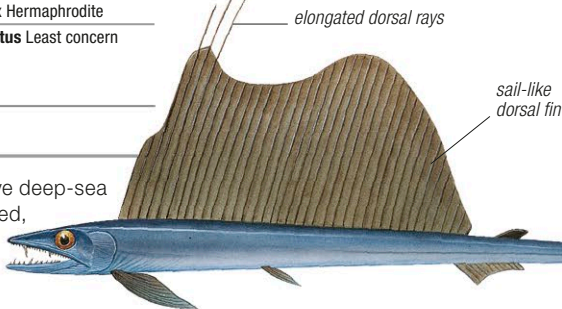


Location Pacific, North Atlantic, Mediterranean

Length Up to 9¼ ft (2.8 m)
Weight 10 lb (4.5 kg)
Sex Hermaphrodite
Status Least concern



This large and distinctive deep-sea fish has a spindle-shaped, iridescent body, a



deeply forked tail, and a long dorsal fin that stretches for about two-thirds of its length, and is held upright like a sail. The function of this fin is unknown. The large mouth has big, pointed teeth for snagging slippery prey. This fish is a useful source of deep-sea specimens for scientists, because the fish in its stomach are usually intact.

Synodus variegatus

Variegated lizardfish



Location Pacific, Indian Ocean



With its pointed snout and large mouth, this fish's head looks very much like a lizard's, when viewed from the side. It often sits on the seafloor, propped on its pelvic fins, with its head raised in a very lizardlike fashion. It may also bury itself in the sand with only its snout and eyes visible, and lunge at small fishes passing by. The variegated lizardfish has a slender, tubular body and its brown, orange, or reddish coloration effectively camouflages it against the sea floor.

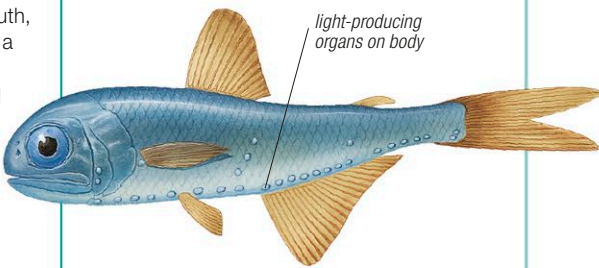


Myctophum asperum

Prickly lanternfish



Location North and South Pacific, W. and E. Atlantic, Indian Ocean



One of the most numerous of the deep-sea fishes and an important link in ocean food chains, lanternfishes eat mainly planktonic crustaceans, and are in turn eaten by seabirds, seals, and larger fishes such as tuna. They derive their name from the light-emitting organs on the head and body. The arrangement of these light organs is an important feature for the identification of the 250 or so species. Differing arrangements of larger light organs near the tail indicate whether the fish is male or female.